1. Artificial Neural Networks (ANN)

- **Definition:** An Artificial Neural Network (neural network) is a computational model designed to mimic the way nerve cells function in the human brain.
- **Structure and Layers:** ANNs contain artificial neurons known as **units**, which are organized into a series of layers.
  - Depending on system complexity, a layer can hold anywhere from a dozen units to millions of units.
  - **Input Layer:** Receives incoming data from the outside world for the network to analyze or learn about.
  - **Hidden Layer(s):** One or multiple layers that transform the inputs into data that is valuable for the output layer.
  - **Output Layer:** Provides the final response or output of the network based on the input provided.

2. Comparing Types of Intelligence

The textbook outlines distinct attributes across human, animal, and machine capabilities:

Core Definitions

- **Human Intelligence:** Gifted by God to humans, it represents a person's scholarly ability to think, learn from experiences, understand complex ideas, solve numerical problems, and adapt to new situations.
- **Animal Intelligence:** The combination of skills and abilities that allow animals to survive, live in, and adapt to their specific natural environments.
- **Artificial Intelligence (AI):** Technology created by humans that mimics human behavior and thinking by processing the information it receives.

Key Differences Between Human and Artificial Intelligence

Feature	Human Intelligence	Artificial Intelligence (AI)
Nature vs. Format	Natural; acts via analogous processing.	Digital; intends to mock human behavior through cognitive procedures.
Processing Method	Uses the brain, memory, and reasoning to think.	Totally dependent on the specific data given to them.
Learning Capability	Learns naturally from past mistakes and intelligent attitudes.	Cannot think freely; learns from information and training but never attains unique human thinking.
Adaptability	Adapts quickly to new environmental changes to learn and ace abilities.	Takes much more time to adjust to structural changes.
Energy Consumption	Highly efficient; human brains run on about **25 watts**.	Less efficient; modern computers typically use about **2 watts** of energy.
Speed & Data Handling	Cannot beat the processing speed of computers.	Handles much larger volumes of data at a vastly speedier rate.
Social & Emotional Intelligence	Excellent at social interaction, sensing emotions, and possessing self-awareness.	Has not fully mastered social cues, emotional elegance, or true self-awareness.

3. Strong AI vs. Weak AI

Strong Artificial Intelligence

- **Definition:** A theoretical form of machine intelligence supporting the view that machines can truly develop human consciousness equal to human beings.
- **Characteristics:** * Can think, mindfully accomplish complex tasks, and make independent decisions without human interference.

    - Algorithms allow them to act fluidly in different dynamic environments.
    - *Note:* There are currently **no proper real-world examples** of Strong AI as it is still in its initial conceptual stage.

Weak Artificial Intelligence (Narrow AI)

- **Definition:** AI with limited functionality designed to accomplish specific, pre-programmed tasks that do not encompass the full spectrum of human cognition.
- **Characteristics:** * Devices cannot think on their own; they are just made to *look* intelligent.
    - Actions are entirely predetermined by algorithms entered by human programmers.
- **Examples:** Google Search Engine, personalized quiz tools, and automated tutoring systems that provide feedback based on a student's performance.

4. Holographic Technology & Mixed Reality (MR)

Holograms & Holography

- **Holography:** The science and technology used to record photographic images in a three-dimensional format using light interference patterns.
- **Hologram:** A three-dimensional (3D) image created using light interference patterns.
    - Unlike traditional 2D images that display flat data, holograms show depth, making objects look real and allowing viewers to see different angles depending on their position.
    - **Creation Process:** Typically involves splitting a coherent light source (like a laser) into two beams: a *reference beam* and an *object beam* (which reflects off the object being captured) to record the intersecting interference pattern onto a material.
- **Real-Life Creation Methods:** Can be created via computer (using augmented reality glasses) or physically (for optical displays).

Mixed Reality (MR / 3D Mixed Reality)

- **Definition:** A blend of real-world environments and digital elements where physical and virtual objects interact in real time.
- **Features:** Users can see both worlds simultaneously. Virtual 3D elements/holograms can be anchored onto real physical tables and manipulated by the user.
- **Hardware Devices:** Requires specialized headsets like **Microsoft HoloLens** or **Magic Leap**, or optimized smartphones.

Applications of Holography

1. **Medical Imaging:** Used to create highly accurate 3D images of organs and tissues to assist with medical diagnostics and surgical planning.
2. **Art and Entertainment:** Used in unique visual displays, exhibitions, and concert performances featuring virtual performers.
3. **Security and Anti-Counterfeiting:** Frequently applied to credit cards, official identification cards, and paper currency to prevent duplication.
4. **Data Storage:** Holographic data storage is an emerging technology that records data in 3D space, leading to vastly higher data densities than traditional methods.
5. **Industry & General MR Uses:** Used for architectural design visualization, immersive gaming, remote collaboration, and industrial maintenance/repair instructions.